

DRY-RUN GUIDE | armstrong.js

Source: javascript\2nd_belt\armstrong.js

CORE BEHAVIOR

Reads the first whitespace token as text, raises each digit to the token length, sums the powers, and compares the sum to the parsed integer.

REPRESENTATIVE INPUT / CALL

Input: '153'.

STEP TRACE

Token length=3; $1^3+5^3+3^3=153$; strict equality succeeds.

EXPECTED OUTPUT / RETURN

'Armstrong'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | // Question: Check whether a number is an Armstrong number.
002 | // Add your solution here.
003 |
004 | const fs = require('fs');
005 |
006 | function main() {
007 |     // Read standard input and trim any whitespace
008 |     const input = fs.readFileSync(0, 'utf-8').trim();
009 |     if (!input) return;
010 |
011 |     // Get the number as a string to count its digits
012 |     const numStr = input.split(/\s+/)[0];
013 |     const n = numStr.length;
014 |     const targetNumber = parseInt(numStr, 10);
015 |
016 |     // Calculate the sum of digits raised to the power of n
017 |     let sum = 0;
018 |     for (let i = 0; i < n; i++) {
019 |         sum += Math.pow(parseInt(numStr[i], 10), n);
020 |     }
021 |
022 |     // Check if the sum equals the original number
023 |     if (sum === targetNumber) {
024 |         console.log("Armstrong");
025 |     } else {
026 |         console.log("Not Armstrong");
027 |     }
028 | }
029 |
030 | main();
031 |
```